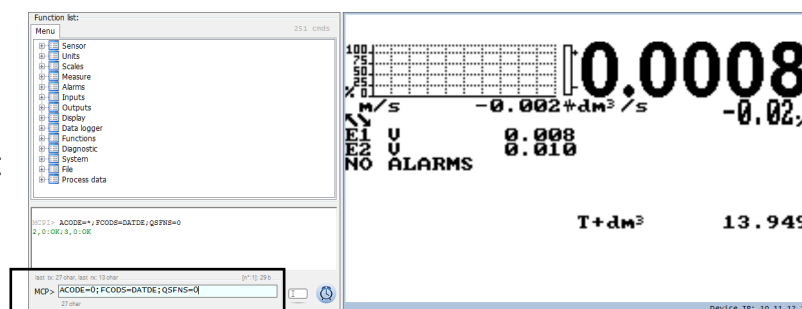


Access CODE	[MCP ONLY]	AL0	[ACODE]
-------------	------------	-----	---------

Input the right access code

Exemple set quick start menù function for mcp.



ACODE=0; FCODS=[MCP COMMAND]; QSFNS=1

Access Code

INSERT MCP Command of the function to be activated in the QS menu.

Quick Start Function Selection

LINK Terminate	[MCP ONLY]	AL0	[LTERM]
----------------	------------	-----	---------

Terminate the PPP data link

MCPI session QUIT	[MCP ONLY]	AL0	[MQUIT]
-------------------	------------	-----	---------

Quit the MCPI connection

Functions LIST	[MCP ONLY]	AL0	[FLIST]
----------------	------------	-----	---------

View list of all available converter functions.

Functions LIST Compact	[MCP ONLY]	AL0	[FLISC]
------------------------	------------	-----	---------

View compact list of all available converter functions.

Functions Menu SElection	[MCP ONLY]	AL0	[FMSEL]
--------------------------	------------	-----	---------

Select menu for functions list

ConFfiguration LiST	[MCP ONLY]	AL0	[CFLST]
---------------------	------------	-----	---------

Configuration parameter list. The list with the status / values of the converter parameter.

Volume Totalizer Total Positive Set	[MCP ONLY]	AL4	[VTTPS]
-------------------------------------	------------	-----	---------

Totaliz.T+ value set

Volume Totalizer Partial Positive Set	[MCP ONLY]	AL4	[VTPPS]
---------------------------------------	------------	-----	---------

Totaliz.P+ value set

<i>Volume Totalizer Total Negative Set</i>	[MCP ONLY]	AL4	[VTTNS]
Totaliz.T- value set			
<i>Volume Totalizer Partial Negative Set</i>	[MCP ONLY]	AL4	[VTPNS]
Totaliz.P- value set			
<i>Volume Total Positive Overflow Set</i>	[MCP ONLY]	AL4	[VTPOS]
Totaliz.T+ overflow value set			
<i>Volume Partial Positive Overflow Set</i>	[MCP ONLY]	AL4	[VPPOS]
Totaliz.P+ overflow value set			
<i>Volume Total Negative Overflow Set</i>	[MCP ONLY]	AL4	[VTNOS]
Totaliz.T- overflow value set			
<i>Volume Partial Negative Overflow Set</i>	[MCP ONLY]	AL4	[VPNOS]
Totaliz.P- overflow value set			
<i>CPU MaX.recorded temperature</i>	[MCP ONLY]	AL6	[CPUMX]
CPU max.recorded temperature			
<i>CPU MiN.recorded temperature</i>	[MCP ONLY]	AL6	[CPUMN]
CPU min.recorded temperature			
<i>Board T1 MaX.recorded temperature</i>	[MCP ONLY]	AL6	[BT1MX]
T1 max.recorded temperature			
<i>Board T1 MiN.recorded temperature</i>	[MCP ONLY]	AL6	[BT1MN]
T1 min.recorded temperature			
<i>Board T2 MaX.recorded temperature</i>	[MCP ONLY]	AL6	[BT2MX]
T2 max.recorded temperature			
<i>Board T2 MiN.recorded temperature</i>	[MCP ONLY]	AL6	[BT2MN]
T2 min.recorded temperature			
<i>Calibration OFset Register 0</i>	[MCP ONLY]	AL6	[COFR0]
Calibration offset register 0			
<i>Calibration OFset Register 1</i>	[MCP ONLY]	AL6	[COFR1]
Calibration offset register 1			

<i>Calibration GAIN Register 0</i>	[MCP ONLY]	<i>AL6</i>	[CGAR0]
---	-------------------	-------------------	----------------

Calibration gain register 0

<i>Calibration GAIN Register 1</i>	[MCP ONLY]	<i>AL6</i>	[CGAR1]
---	-------------------	-------------------	----------------

Calibration gain register 1

<i>Calibration GAIN Register 2</i>	[MCP ONLY]	<i>AL6</i>	[CGAR2]
---	-------------------	-------------------	----------------

Calibration gain register 2

<i>Calibration GAIN Register 3</i>	[MCP ONLY]	<i>AL6</i>	[CGAR3]
---	-------------------	-------------------	----------------

Calibration gain register 3

<i>Calibration GAIN Register C</i>	[MCP ONLY]	<i>AL6</i>	[CGARC]
---	-------------------	-------------------	----------------

Calibration gain register C

MENU 14 - FILE (ONLY MCP)

<i>File Transfer ABoRt</i>	[MCP ONLY]	AL2	[FTABR]
Abort the current File Transfer			
<i>File Transfer STAtE</i>	[MCP ONLY]	AL0	[FTSTA]
Show the File Transfer state			
<i>Read Last EVenTs</i>	[MCP ONLY]	AL2	[RLEVT]
Read the latest system events			
<i>Read All EVenTs</i>	[MCP ONLY]	AL2	[RAEVT]
Read all current system events			
<i>Read Last Logged DaTa</i>	[MCP ONLY]	AL2	[RLLDT]
Read the latest logged data			
<i>Read All Logged DaTa</i>	[MCP ONLY]	AL2	[RLLDT]
Read all current logged data			
<i>Read Last Sensor Verify Data</i>	[MCP ONLY]	AL2	[RLSVD]
Read the latest sensor ver. data			
<i>Read All Sensor Verify Data</i>	[MCP ONLY]	AL2	[RASVD]
Read all sensor verify data			
<i>File SEND</i>	[MCP ONLY]	AL2	[FSEND]
Set file name for read operation			
<i>File ReCeIvE</i>	[MCP ONLY]	AL5	[FRCVE]
Set file name for write operation			
<i>File ReCeive APpend mode</i>	[MCP ONLY]	AL5	[FRCAP]
Set file name for write-append			
<i>File OFFSet position</i>	[MCP ONLY]	AL2	[FOFFS]
Set file offset position			
<i>ConFiGuration file WRite</i>	[MCP ONLY]	AL2	[CFGWR]
Save the configuration to a file			
<i>ConFiGuration file ReaD</i>	[MCP ONLY]	AL2	[CFGRD]
Read the configuration from file			

<i>FuNction list file WRite</i>	[MCP ONLY]	<i>AL2</i>	[FNCWR]
--	-------------------	-------------------	----------------

Save the functions list to file

<i>Function Enable Status WRite</i>	[MCP ONLY]	<i>AL6</i>	[FESWR]
--	-------------------	-------------------	----------------

Save function enable status to file

<i>Quick Start function Status WRite</i>	[MCP ONLY]	<i>AL6</i>	[QSSWR]
---	-------------------	-------------------	----------------

Save quick start function enable.

MENU 15 - PROCESS DATA (ONLY MCP)

<i>OUTput 1 Set</i>	[MCP ONLY]	<i>AL0</i>	[OUT1S]
Set value for digital output 1			
<i>OUTput 2 Set</i>	[MCP ONLY]	<i>AL0</i>	[OUT2S]
Set value for digital output 2			
<i>Digital INput 1 Status</i>	[MCP ONLY]	<i>AL0</i>	[DIN1S]
Digital input 1 status read			
<i>Flow Rate Full Scale in chosen Units</i>	[MCP ONLY]	<i>AL0</i>	[FRFSU]
F.rate f.scale in chosen units			
<i>Flow Rate Value PerCentage</i>	[MCP ONLY]	<i>AL0</i>	[FRVPC]
Flow rate value in percentage			
<i>Flow Rate Value Percentage without cut-off</i>	[MCP ONLY]	<i>AL0</i>	[FRVPX]
F.rate in perc.without cut-off			
<i>Flow Rate Value Binary without cut-off</i>	[MCP ONLY]	<i>AL0</i>	[FRVBX]
F.rate in binary.without cut-off			
<i>Flow Rate Value Technical Unit</i>	[MCP ONLY]	<i>AL0</i>	[FRVTU]
F.rate value in unit of measure			
<i>Volume Totalizer Total Positive Value</i>	[MCP ONLY]	<i>AL0</i>	[VTTTPV]
Totaliz.T+ read value			
<i>Volume Totalizer Partial Positive Value</i>	[MCP ONLY]	<i>AL0</i>	[VTPPV]
Totaliz.P+ read value			
<i>Volume Totalizer Total Negative Value</i>	[MCP ONLY]	<i>AL0</i>	[VTTNV]
Totaliz.T- read value			
<i>Volume Totalizer Partial Negative Value</i>	[MCP ONLY]	<i>AL0</i>	[VTPNV]
Totaliz.P- read value			
<i>Volume Totalizer Total Positive Overflow</i>	[MCP ONLY]	<i>AL0</i>	[VTTPO]
Totaliz.T+ number of overflows			
<i>Volume Totalizer Partial Positive Overflow</i>	[MCP ONLY]	<i>AL0</i>	[VTPPO]
Totaliz.P+ number of overflows			
<i>Volume Totalizer Total Negative Overflow</i>	[MCP ONLY]	<i>AL0</i>	[VTTNO]
Totaliz.T- number of overflows			

<i>Volume Totalizer Partial Negative Overflow</i>	[MCP ONLY]	<i>AL0</i>	[VTPNO]
Totaliz.P- number of overflows			
<i>Board TeMPeratures</i>	[MCP ONLY]	<i>AL0</i>	[BTMPS]
Board temperatures			
<i>CPU temperature</i>	[MCP ONLY]	<i>AL0</i>	[CPUTP]
CPU temperature			
<i>Sensor CoiLs TemPerature</i>	[MCP ONLY]	<i>AL0</i>	[SCLTP]
sensor's coils temperature			
<i>LiQuid VELocity</i>	[MCP ONLY]	<i>AL0</i>	[LQVEL]
Liquid velocity			
<i>AVeraGe process data Samples Number</i>	[MCP ONLY]	<i>AL0</i>	[AVGSN]
N.of samples for averaged values			
<i>ALARM status</i>	[MCP ONLY]	<i>AL0</i>	[ALARM]
Active alarm(s) status			
<i>Sensor TeSt Result Code</i>	[MCP ONLY]	<i>AL0</i>	[STSRC]
Sensor test result code			
<i>Main power status</i>	[MCP ONLY]	<i>AL0</i>	[MPWRS]
Status of main power supply			
<i>INput RESistance</i>	[MCP ONLY]	<i>AL0</i>	[INRES]
Equivalent Input resistance			
<i>INput VoLtageS</i>	[MCP ONLY]	<i>AL0</i>	[INVLS]
Electrodes input voltages			
<i>System Battery Voltage</i>	[MCP ONLY]	<i>AL0</i>	[SBVLT?]
View battery voltage			
<i>System Battery Charge Status</i>	[MCP ONLY]	<i>AL0</i>	[SBCHS?]
View system battery charge status			
<i>Measure BUFFers</i>	[MCP ONLY]	<i>AL0</i>	[MBUFF]
Measure buffers data read			
<i>SEquence NumBer</i>	[MCP ONLY]	<i>AL0</i>	[SEQNB]
Sequence number			

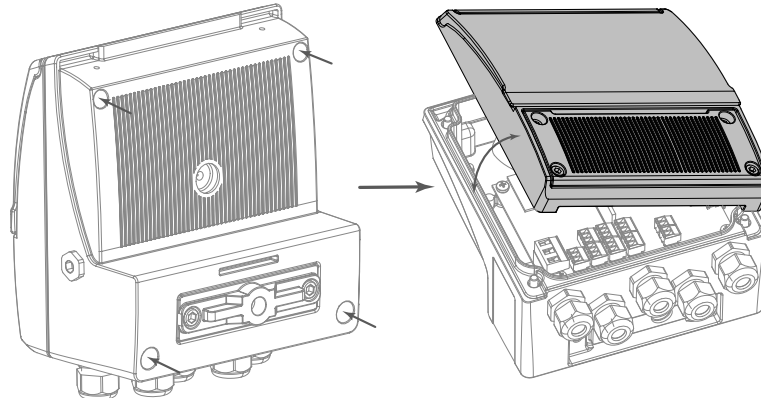
METER DATA

This function allows the import of data from one converter to another up to level 4 included. The hardware configurations and the corresponding calibration values are not restored. The "data import" procedure can be performed **one time only**, since the directory, according to the board's SERIAL NUMBER, will be renamed.

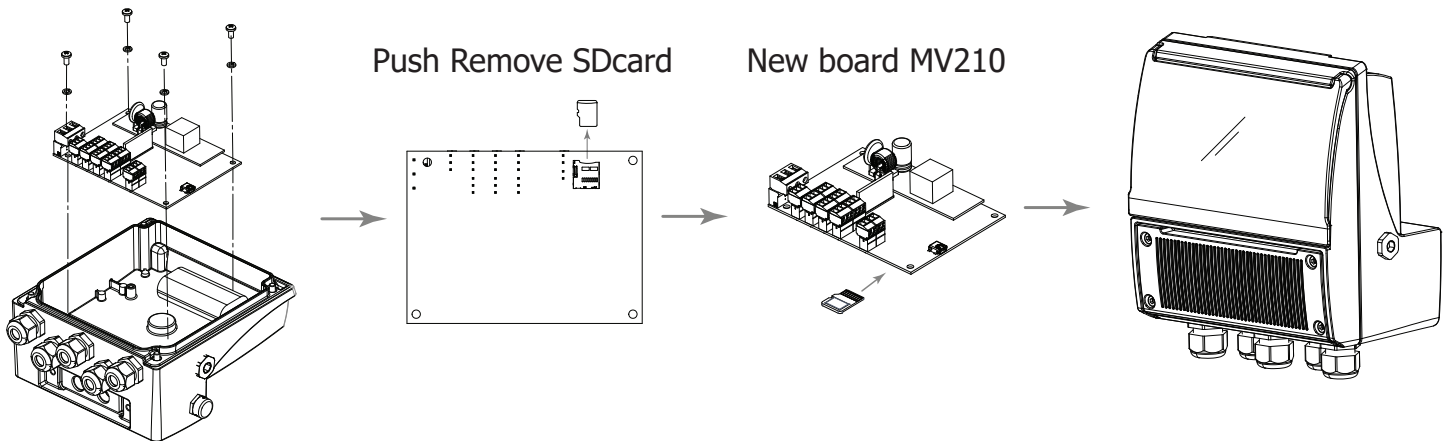
OPERATING PROCEDURE TO CHANGE THE CONVERTER BOARD



Remove the 4 screws (see "MV210 construction" page10) to remove the main housing. Attention of any electrical cables.



Remove the 4 screws (see "MV210 construction" page10) to remove the MV210 board. Attention of any electrical cables. Remove the SD card and insert the new MV210 board.



Assembled the card turn on the converter and enable [Save Conv. F. def.] function to restore data.



View function "11.8" page30.

B.I.V. (BUILT-IN VERIFICATOR)

BIV, abbreviation for Built In Verificator, is available as option for MV210 converters and must be enabled by the manufacturer. It is also necessary that the SD card is activated to store saved data. The analysis of collected data performed by a dedicated IsoBIV software running on another device (PC). The simplicity of test procedures minimize the risk of handling errors; maximum safety and reliability thanks to the traceable factory calibration and internal references complement the safety by design principle with minimal failure rates

IsoBIV allows to create and print a report as validation of device functionality/measure error.

Operation and Conditions of Use.

The system is based on periodic measurements performed every hour or using a manual command (MCP command = SVERC). The sensor parameters are measured and compared with previously measured and stored reference values. Each time the system performs a series of measurements on the sensor and records them in a file called "STESTLOG.CSV", which resides in the main directory of the SD memory of the converter.

The sensor test can also be carried out without the active BIV system, but in this case only the presence of isolation losses and the overall good functioning of the sensor such the coil resistance, the excitation current and the rising times of the current within the generic limits that guarantee operation. Instead if BIV is active, the measurements are deeper and the measured values are tested by comparing them with a set of characteristic sensor parameters measured at the time of installation.

Saving Reference Values (Characteristic Parameters)

After sensor installation, the parameters that will be used as reference for the BIV system and the IsoBIV data analysis software must to be measured.

The characteristic values of the coil circuits are saved in the converter memory at the factory before to ship the instrument. For the reference measure of the electrodes circuits, there is a specific function that perform the measures of voltage and resistance at the installation site. This function is managed ONLY by the IsoBIV program, which through a simple wizard will set the converter to perform the measures in the specific measurement point where the meter is installed.

To activate BIV, these functions must be verified:

- ☐ **[ASVFE=1]** : It enable the sensor's automatic test every hour. The ASVFE function in Menu 1 with access level 3 can be also activated using the instrument's display.

This feature can be enabled even if the SD card is not installed and if the BIV function is not active; in such a case the sensor file is NOT created and any alarms will be generated either if the data deviation from the reference data is outside the sesnor's limits.

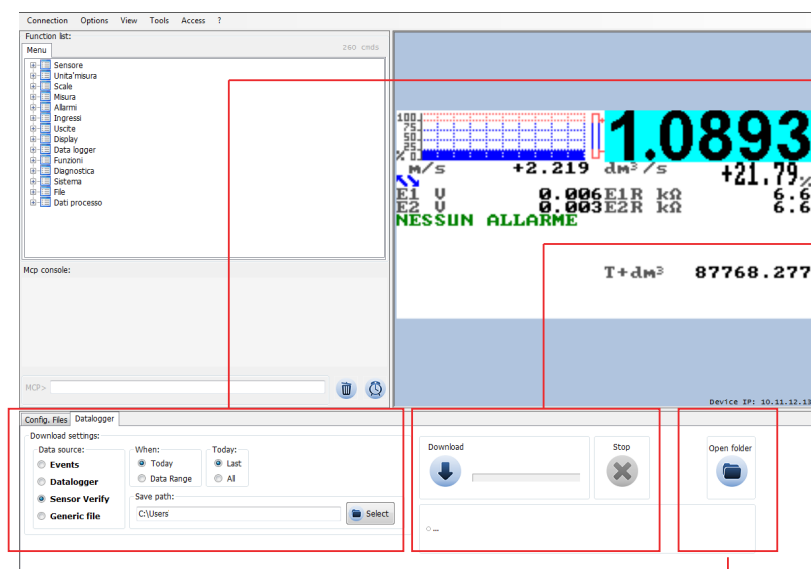
Practically, in the absence of the necessary hardware permissions, this function is useful to test the insulation of the coils.

Opening and reading manually the files STESTLOG.CSV

The list below describes the steps for saving and reading STESTLOG.CSV file.

❑ OPEN INTERFACE STARTING THE PROGRAM MCP 

❑ FOLLOW THE STEPS HERE BELOW



- 1
 - Select Sensor Verify
 - Choose the date or reference period of the which one wants to carry out the evaluation of the
 - Choose the directory where to save the file

- 2
 - Press Download to download the file.
 - Press Download to download the file.
 - At the end of the download the stestlog.csv file will be automatically opened by the program

Search for the consultation of the files inside of the drive SD card.

Path structure for searching the files in the SD card of the converter

